

H162. Where Flood Connects Us: Integrating Physical Models, Social Dimensions, and AI Innovations

Title: Adaptive Chaos Control with Deep Learning of Atmospheric Dynamics to leverage the power of the atmospheric circulation with nudges to move or defuse storms and avert catastrophic flooding

Atmospheric Rivers, Tropical and Extratropical Cyclones, Nor'easters, and Large-Scale Frontal Storms that interact with Mesoscale Convective Complexes are examples of weather extremes that can lead to catastrophic flooding. Our ability to predict the trajectory and impacts of these systems has improved dramatically over the last few decades, thanks to better models and, in large part, to data availability and assimilation. Yet, significant challenges remain, as the space and time specificity of the intensity and duration of the associated storm precipitation is limited, especially at longer lead times. The recent flooding in Texas raised questions as to whether there was adequate warning of flooding, and also whether people would have responded if the warning had been effectively communicated much earlier, giving them enough time to organize and evacuate. This question arises nearly every time such an event happens, unless there is a government-ordered evacuation.

The inability to predict the details of the storms at longer lead times connects to the well-demonstrated chaotic nature of atmospheric circulation dynamics. The sensitivity to initial conditions leads to the amplification of small perturbations, or uncertainty in initial conditions, which renders long-range forecasts unreliable unless they are updated with continuous data assimilation. The Lorenz butterfly effect is the iconic representation of this paradigm.

Weather Jiu Jitsu is a paradigm we have introduced to represent the opportunity to harness the power of nature to change the trajectory of atmospheric circulation via appropriately timed and located perturbations or nudges. In this way, catastrophic flooding events or storms can be diffused by spreading storms over larger areas and/or diverted so that their impacts are minimized.

If successful, Weather Jiu Jitsu would provide humanity with a new tool beyond dams and levees, which have been demonstrated to have negative socio-environmental consequences, and beyond insurance mechanisms, which are only exercised by a fraction of the population and are increasingly under stress as the climate changes.

We provide examples of Weather Jiu Jitsu using idealized models and deep learning emulators of weather dynamics, and discuss the opportunity to collaborate on building this new science with modern computational tools for delivering the nudges.